

LOGO

Airline Contrail Impact Report 2024

What are Contrails?

Contrails — the thin, white lines you sometimes see behind airplanes — have a surprisingly large impact on our climate. Contrails warm the planet because contrail clouds act like a blanket on Earth and have a net heating effect. The 2022 IPCC report noted that clouds created by contrails account for roughly 35% of aviation's global warming impact — over half the impact of the world's jet fuel. Find more info about contrails and the climate on our website contrails.org.

What is Global Warming Potential (GWP)?

GWP measures how much warming contrails cause over a number of years compared to CO2. Contrails heat the Earth quickly but for a short time, and GWP helps compare their short-term impact to the longer-lasting greenhouse gas, CO2.

In this report we initially show the contrail impact in CO2e over 20, 50 and 100 years to align with the guidelines from the EU Non-CO2 MRV report starting in 2025. Wherever we only show one value for CO2e we use the middle value, GWP50, as default.

Impact Data

Based on our prediction model, 5.5 Million km (55,501 flight hours) or 4.4% of all [Airline] flights generated warming contrails in 2024.

of Flights ⓘ

15,380 flights

Flight hours ⓘ

55,010 hours

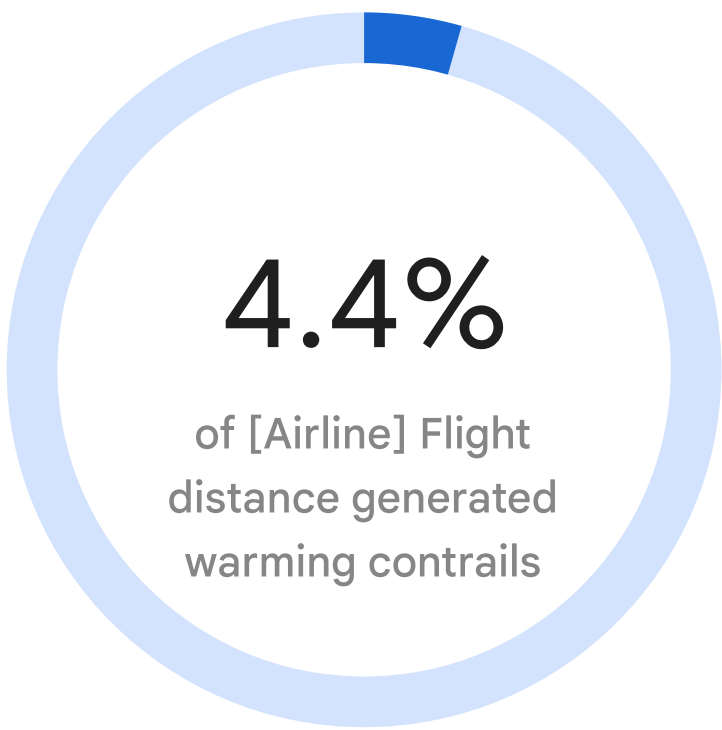
Contrails (GWP 50) ⓘ

275k metric tons CO2e

Fuel burn ⓘ

954k metric tons CO2

What percentage of [Ariline] flights created warming contrails?



- Distance (km) with warming contrails
- Distance (km) without warming contrails

How many flight kilometers of warming contrails?

Flight kilometers for all flights ⓘ

46M km



Flight kilometers creating warming contrails ⓘ

5.5M km



- Nighttime
- Daytime